

Lab 8 - Report

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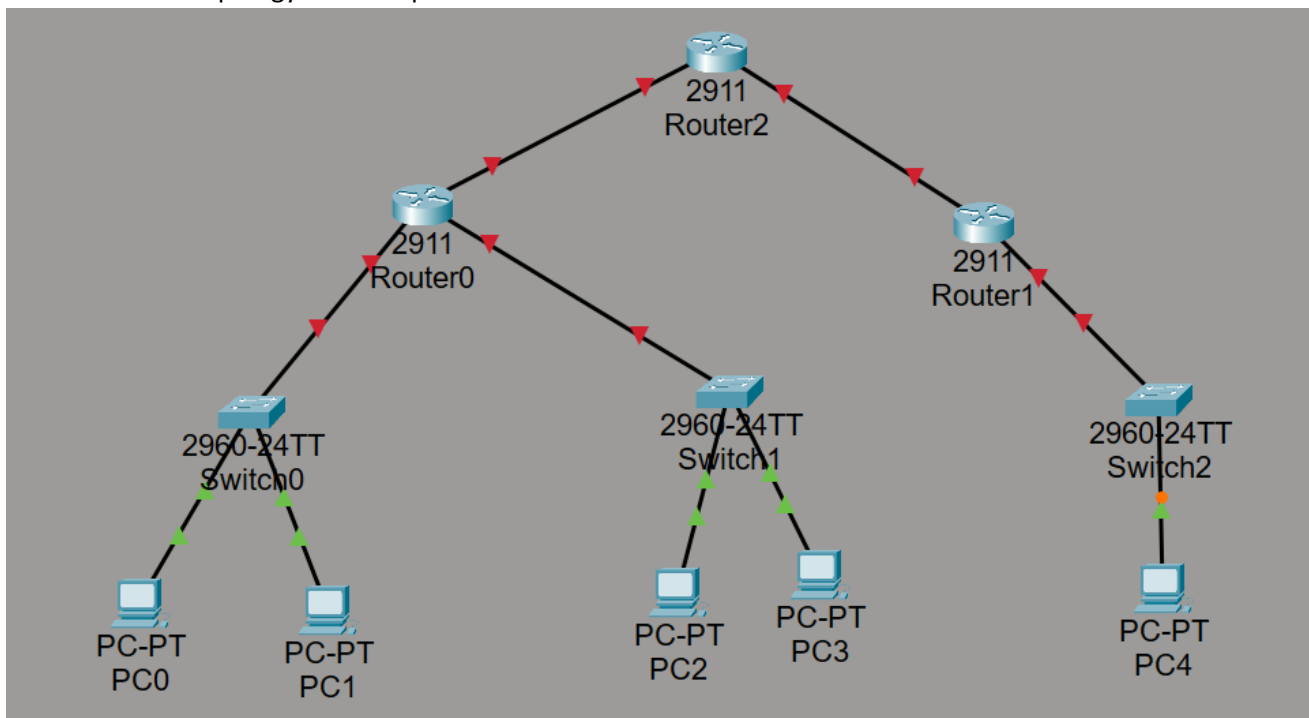
Student ID: 24B031692

Group: Mon 11:00-13:00

Static Routing

Part 1: Network Setup

1.1 Topology and Components



1.2 IP Address Configuration

Router0

Physical Config CLI Attributes

```
Router(config)#int gi
Router(config)#int gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#int gi
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi
Router(config)#int gigabitEthernet 0/2
Router(config-if)#ip add
Router(config-if)#ip address 10.0.0.1
% Incomplete command.
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
```

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Router1

Physical Config CLI Attributes

```
Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi
Router(config)#int gigabitEthernet 0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

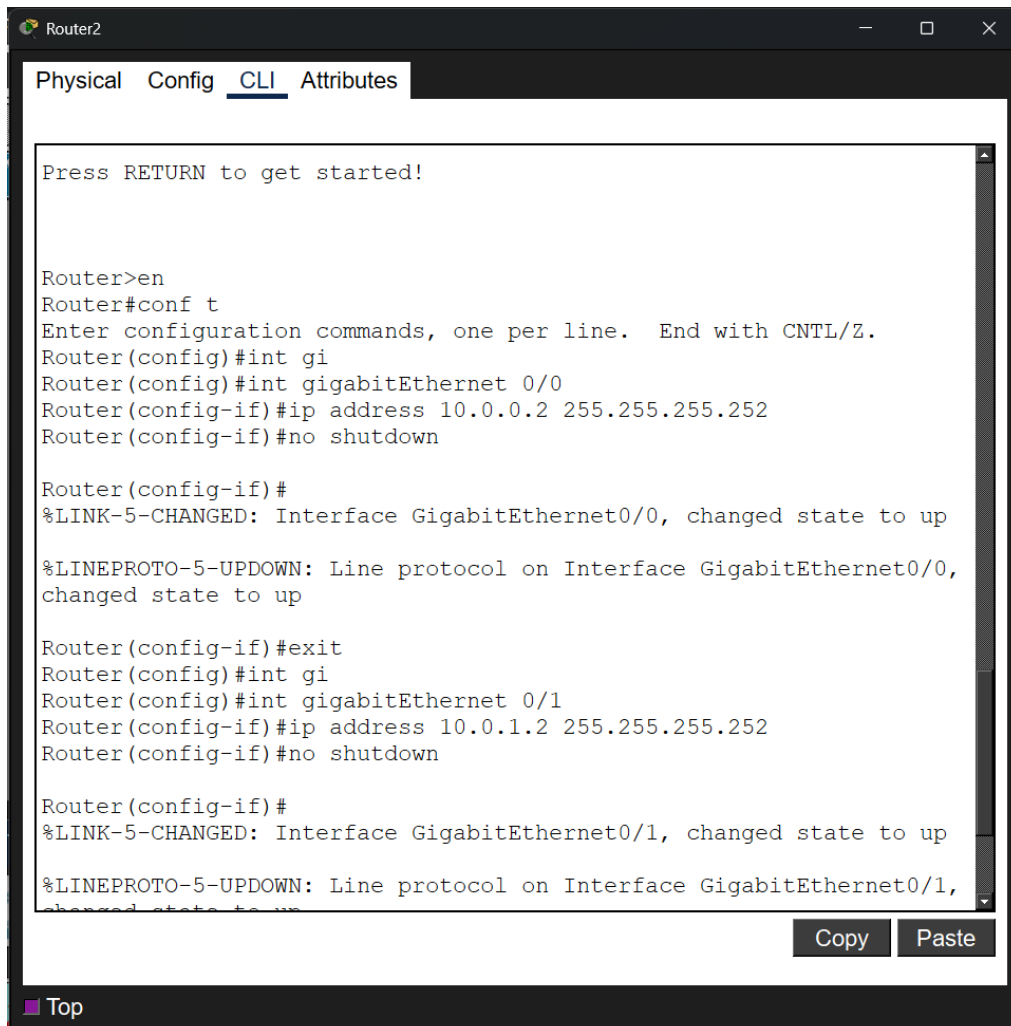
Router(config-if)#exit
Router(config)#int gi
Router(config)#int gigabitEthernet 0/1
Router(config-if)#ip address 10.0.1.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

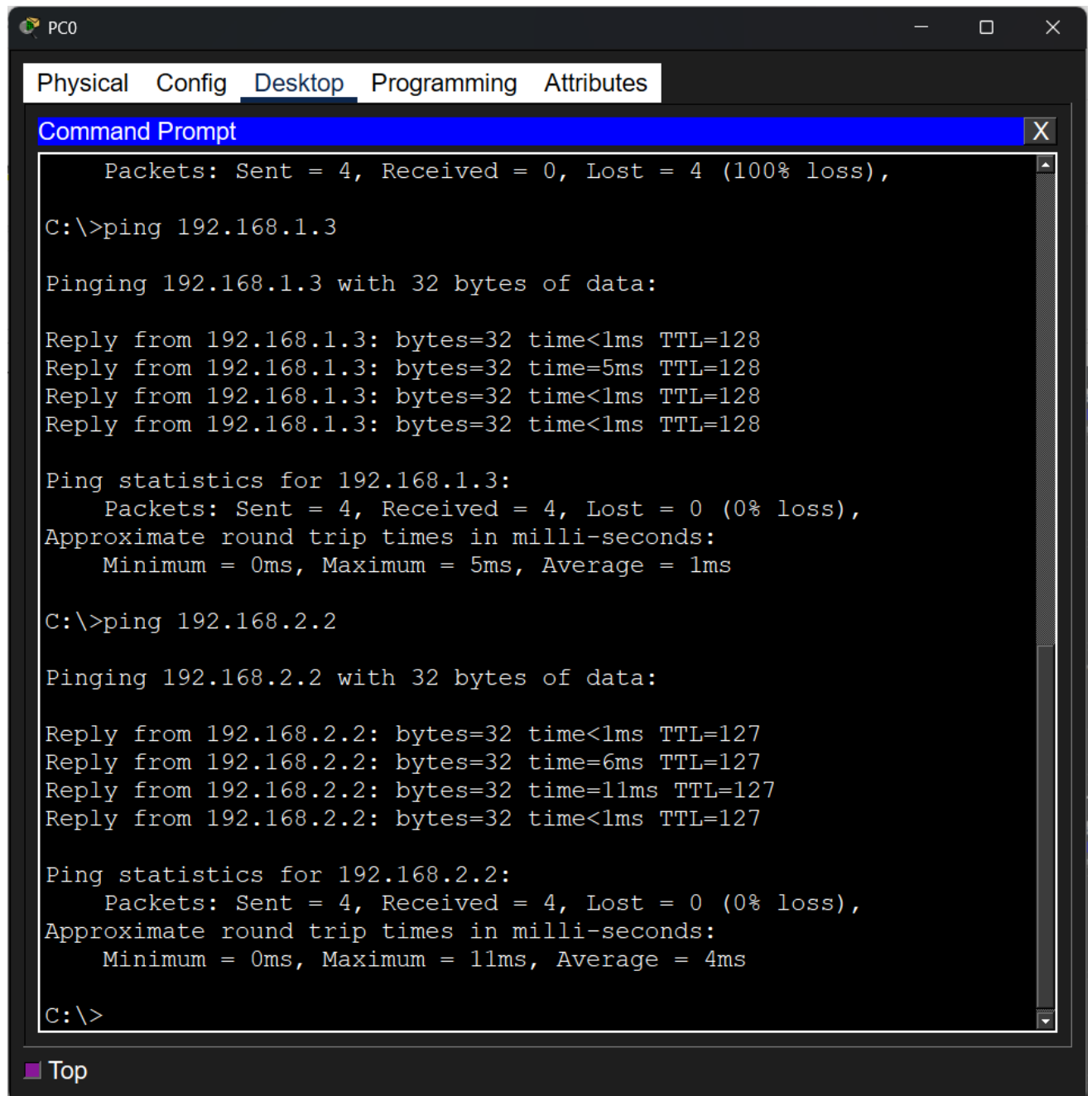
Router(config-if)#end
```

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Step 4: Test the connectivity from PC0 to PC1, Router0 and PC2 using ping command. Was it possible to ping PC2? Why?



The screenshot shows a Windows-style window titled "PC0" with a tabbed interface. The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of two ping commands. The first command is "C:\>ping 192.168.1.3", which shows four successful replies with 0% loss. The second command is "C:\>ping 192.168.2.2", which also shows four successful replies with 0% loss. The window has a "Top" button at the bottom left.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=5ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=6ms TTL=127
Reply from 192.168.2.2: bytes=32 time=11ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms
C:\>
Top
```

Yes. Because both PC are **directly connected** to Router0

1.2.2 Assign IP Addresses to PCs

PC0

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::210:11FF:FED0:A9E5

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC1

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BAFF:FE0D:2DBD

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:42FF:FE2A:E252

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:D3FF:FE6C:6259

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address

192.168.3.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.3.1

DNS Server

0.0.0.0

IPv6 Configuration

☒ Automatic

☒ Static

IPv6 Address

 /

Link Local Address

FE80::202:16FF:FED3:5224

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication

MD5

Username

Password

Top

Part 2: Static Routing Configuration

2.1 Configure Static Routes

Router1

Physical Config CLI Attributes

Press RETURN to get started.

```
Router>conf t
      ^
% Invalid input detected at '^' marker.

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.1.2
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.1.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

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Top

Router0

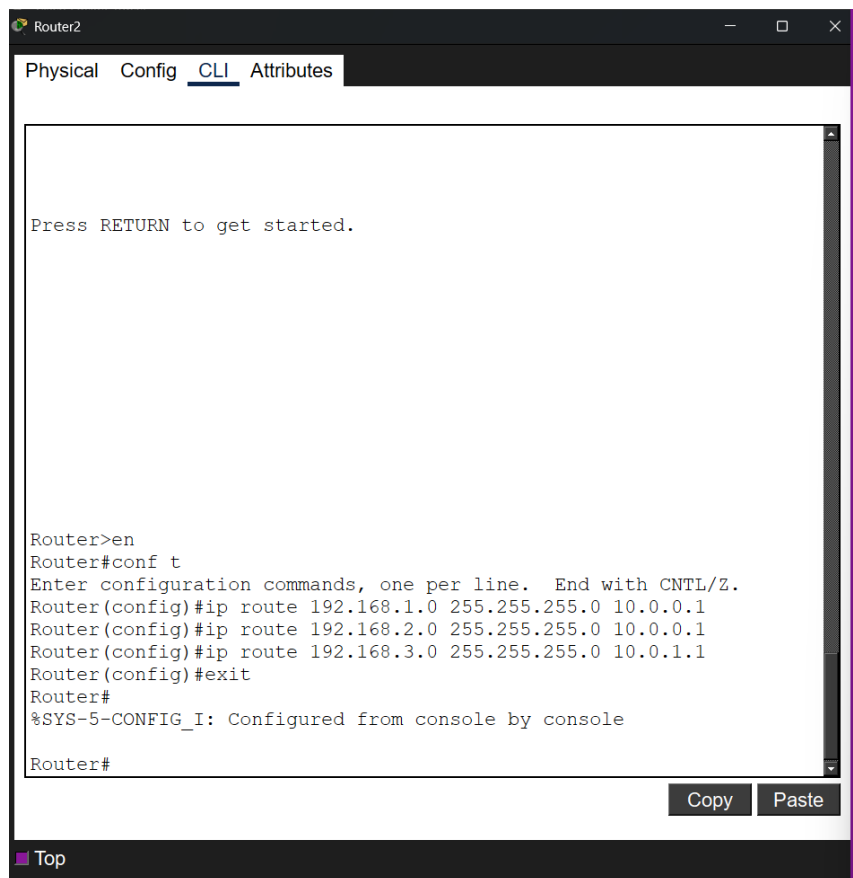
Physical Config CLI Attributes

Press RETURN to get started.

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

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2.2 Configure Default Routes

```
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2  
Router(config)#
```

```
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.1.2  
Router(config)#
```

Part 3: Testing and Verification

3.1 Verify Routing Tables

Router0

Physical Config CLI Attributes

```
%SYS-5-CONFIG_1: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
S       192.168.3.0/24 [1/0] via 10.0.0.2
S*      0.0.0.0/0 [1/0] via 10.0.0.2

Router#
```

Copy Paste

Top

Router1

Physical Config CLI Attributes

```
Co
% Ambiguous command: "co"
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is 10.0.1.2 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.1/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 10.0.1.2
S       192.168.2.0/24 [1/0] via 10.0.1.2
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
S*      0.0.0.0/0 [1/0] via 10.0.1.2

Router#
```

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Top

```
Router2
Physical Config CLI Attributes
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is not set

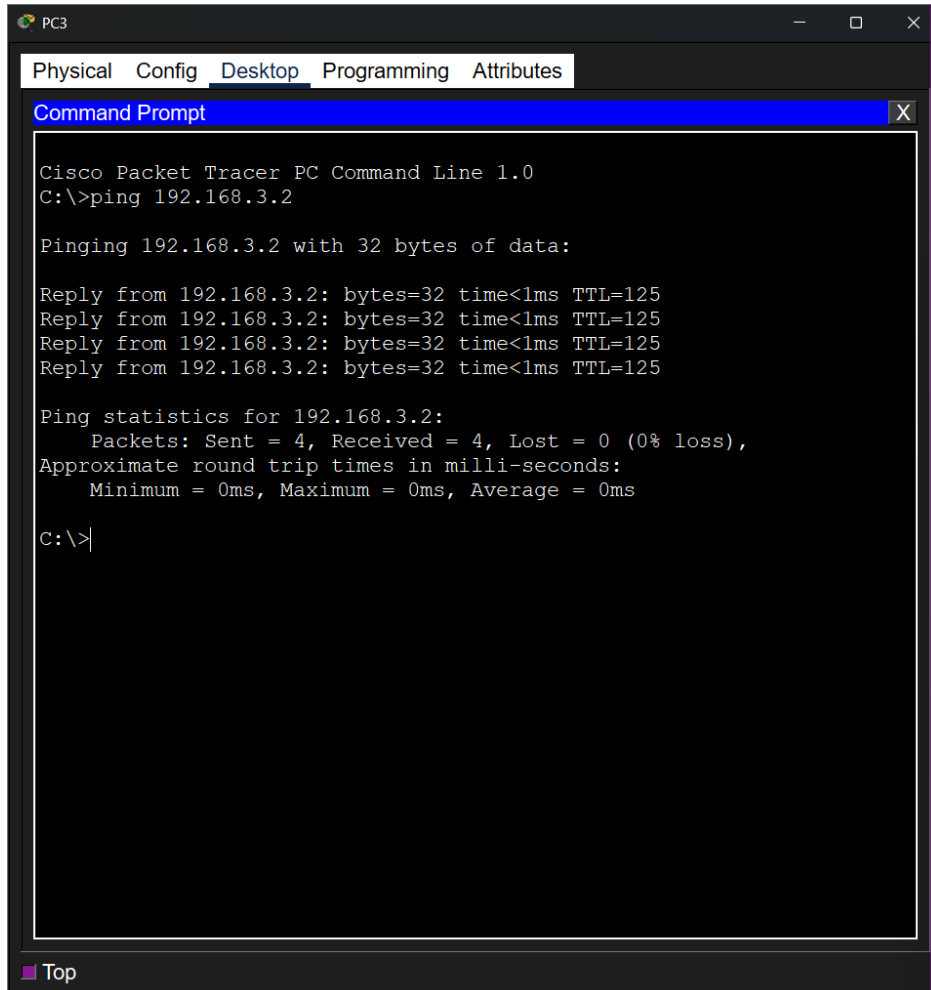
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.2/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 10.0.0.1
S       192.168.2.0/24 [1/0] via 10.0.0.1
S       192.168.3.0/24 [1/0] via 10.0.1.1

Router#
```

Note the difference between the routing tables of Router2 and Router0/Router1. What does “Gateway of last resort is not set” indicate? How can you configure it?

Router0/1 has a default gateway on Router2, and Router2 is a Backbone, so it knows all the routes and has no place to forward "unknown" packets. If it receives a packet with an address not in its table, it simply drops it. How to configure: You need to enter the command: iproute

3.2 Test Connectivity with Ping



PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

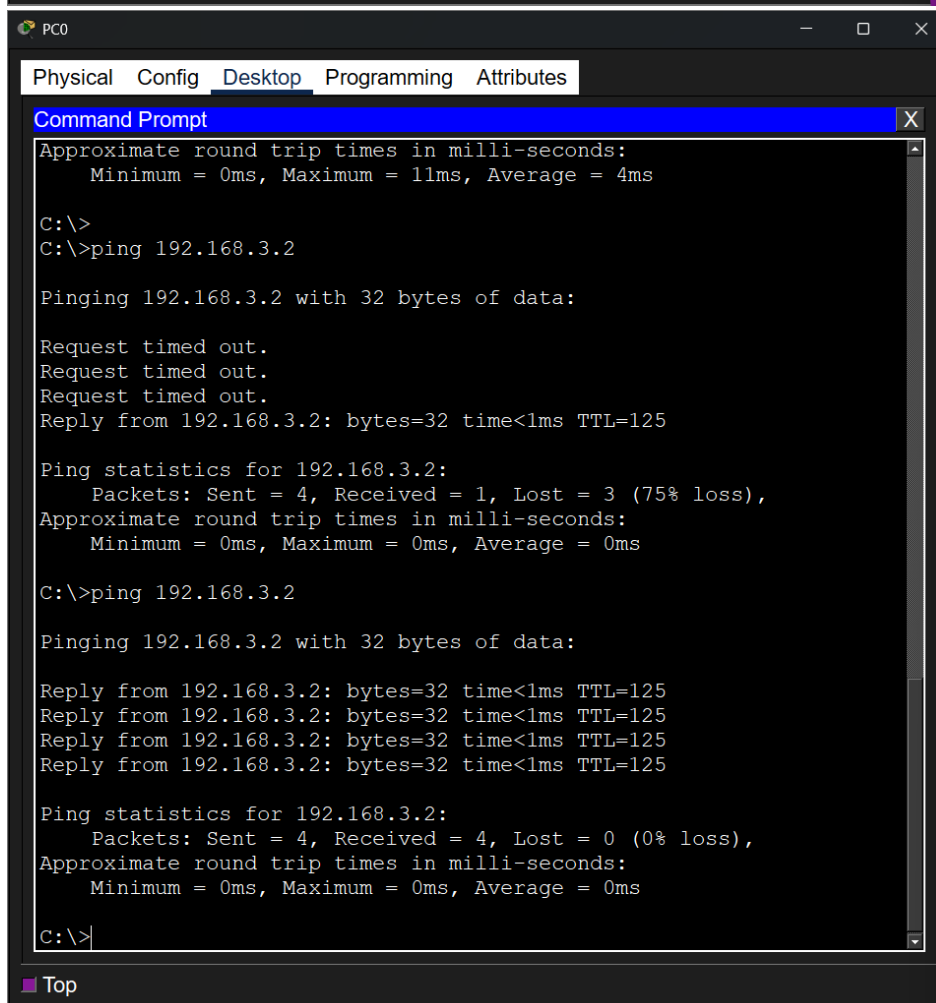
Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Top



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

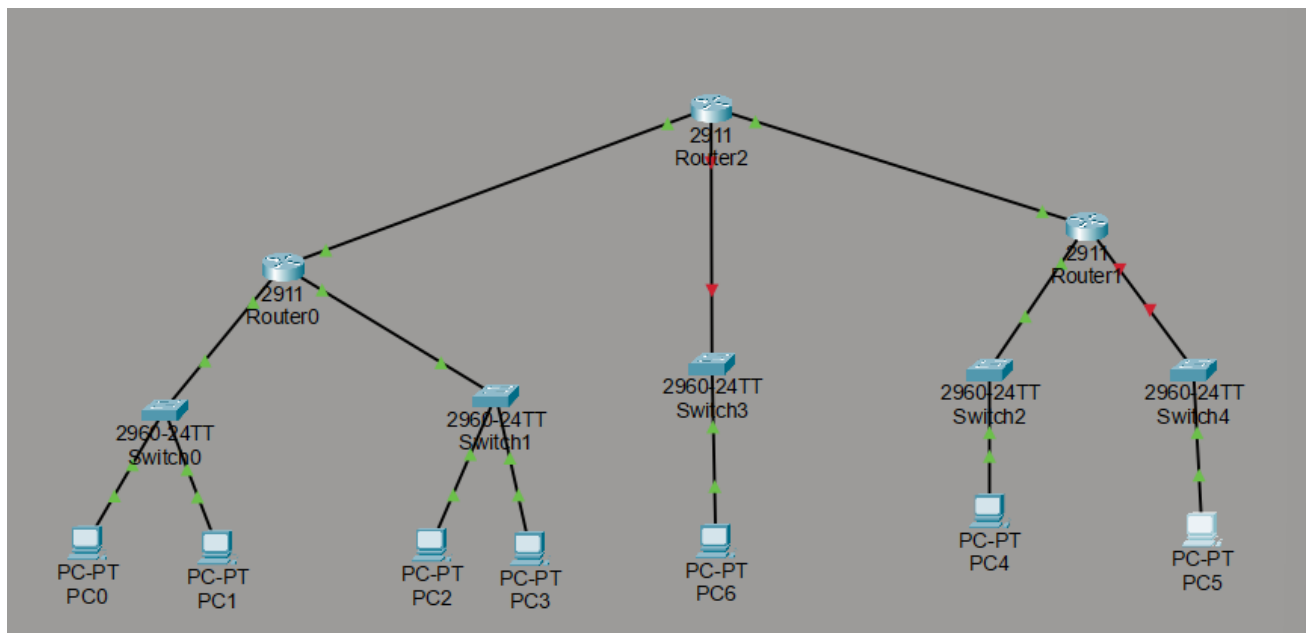
Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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Part 4: Additional Task - Adding a New Subnet

1. Add a new PC5 and connect it to Router1 via a new switch and PC6 to Router2 via a new switch too.



2. Choose an appropriate IP address range for these new subnets (e.g., 192.168.4.0/24).

- Range 192.168.4.0/24 for PC5 and 192.168.5.0/24 for PC6.

3,4. Assign an IP address to the new router interface and configure the new subnet on Router1

Repeat step 3 for Router2.

```
Router1
Physical Config CLI Attributes

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi
Router(config)#int gigabitEthernet 0/2
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#

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Router2
Physical Config CLI Attributes

Router>en
Router#conf
Configuring from terminal, memory, or network [terminal]? t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi
Router(config)#int gigabitEthernet 0/2
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#

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Top
```

5,6. Assign an IP address and default gateway to PC5. Repeat step 5 for PC6.

PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.4.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.4.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C7FF:FE5B:44C5

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC6

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.5.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.5.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:B0FF:FE28:B305

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

7. Configure static routes on Router0, Router1 and Router2 to allow full communication with the new subnet

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.4.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.5.0 255.255.255.0 10.0.0.2
Router(config)#
```

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```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.5.0 255.255.255.0 10.0.1.2
Router(config)#
```

```
Router(config)#ip route 192.168.4.0 255.255.255.0 10.0.1.1
Router(config)#S
```

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8. Verify connectivity by testing the routing tables and using the ping command.

Router0

Physical Config CLI Attributes

```
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
S       192.168.3.0/24 [1/0] via 10.0.0.2
S       192.168.4.0/24 [1/0] via 10.0.0.2
S       192.168.5.0/24 [1/0] via 10.0.0.2
--More--
```

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Top

Router1

Physical Config CLI Attributes

```
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is 10.0.1.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.1/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 10.0.1.2
S       192.168.2.0/24 [1/0] via 10.0.1.2
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/2
L       192.168.4.1/32 is directly connected, GigabitEthernet0/2
S       192.168.5.0/24 [1/0] via 10.0.1.2
--More--
```

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Router2

Physical Config CLI Attributes

```
%SYS-5-CONFIG_1: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.2/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 10.0.0.1
S       192.168.2.0/24 [1/0] via 10.0.0.1
S       192.168.3.0/24 [1/0] via 10.0.1.1
S       192.168.4.0/24 [1/0] via 10.0.1.1
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.5.0/24 is directly connected, GigabitEthernet0/2
L       192.168.5.1/32 is directly connected, GigabitEthernet0/2
--More--
```

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PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time<1ms TTL=125
Reply from 192.168.4.2: bytes=32 time<1ms TTL=125
Reply from 192.168.4.2: bytes=32 time<1ms TTL=125
Reply from 192.168.4.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.4.2:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time<1ms TTL=126
Reply from 192.168.5.2: bytes=32 time<1ms TTL=126
Reply from 192.168.5.2: bytes=32 time<1ms TTL=126
Reply from 192.168.5.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.5.2:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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Part 5: Dynamic Routing with OSPF

Router 0 Configuration

Router0

Physical Config CLI Attributes

```
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/1
L    192.168.2.1/32 is directly connected, GigabitEthernet0/1
S    192.168.3.0/24 [1/0] via 10.0.0.2
S    192.168.4.0/24 [1/0] via 10.0.0.2
S    192.168.5.0/24 [1/0] via 10.0.0.2
S*   0.0.0.0/0 [1/0] via 10.0.0.2

Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip route 192.168.3.0 255.255.255.0 10.0.0.2
Router(config)#no ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

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Router 1 Configuration

Router1

Physical Config CLI Attributes

```
S    192.168.2.0/24 [1/0] via 10.0.1.2
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, GigabitEthernet0/0
L    192.168.3.1/32 is directly connected, GigabitEthernet0/0
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected, GigabitEthernet0/2
L    192.168.4.1/32 is directly connected, GigabitEthernet0/2
S    192.168.5.0/24 [1/0] via 10.0.1.2
S*   0.0.0.0/0 [1/0] via 10.0.1.2

Router#
Router#
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip route 192.168.1.0 255.255.255.0 10.0.1.2
Router(config)#no ip route 192.168.2.0 255.255.255.0 10.0.1.2
Router(config)#no ip route 0.0.0.0 0.0.0.0 10.0.1.2
Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 10.0.1.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

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Router 2 Configuration

Router2

Physical Config CLI Attributes

L 192.168.5.1/32 is directly connected, GigabitEthernet0/2

Router#

Router#

Router#

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#no ip route 192.168.1.0 255.255.255.0 10.0.0.1

Router(config)#no ip route 192.168.2.0 255.255.255.0 10.0.0.1

Router(config)#no ip route 192.168.3.0 255.255.255.0 10.0.1.1

Router(config)#router ospf 1

Router(config-router)#router-id 3.3.3.3

Router(config-router)#network 10.0.0.0 0.0.0.3 area 0

Router(config-router)#network 10.0.1.0 0.0.0.3 area 0

Router(config-router)#

Router(config-router)#ex

01:16:21: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Don

Router(config-router)#exit

Router(config)#

01:16:28: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem

Building configuration...

[OK]

Router#S

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5.4 Verification

Router2

Physical Config CLI Attributes

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

C 10.0.0.0/30 is directly connected, GigabitEthernet0/0

L 10.0.0.2/32 is directly connected, GigabitEthernet0/0

C 10.0.1.0/30 is directly connected, GigabitEthernet0/1

L 10.0.1.2/32 is directly connected, GigabitEthernet0/1

O 192.168.1.0/24 [110/2] via 10.0.0.1, 00:01:52, GigabitEthernet0/0

O 192.168.2.0/24 [110/2] via 10.0.0.1, 00:01:52, GigabitEthernet0/0

O 192.168.3.0/24 [110/2] via 10.0.1.1, 00:01:42, GigabitEthernet0/1

S 192.168.4.0/24 [1/0] via 10.0.1.1

192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.5.0/24 is directly connected, GigabitEthernet0/2

L 192.168.5.1/32 is directly connected, GigabitEthernet0/2

Router#

Router#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address
Interface				
2.2.2.2	1	FULL/DR	00:00:30	10.0.1.1
GigabitEthernet0/1				
1.1.1.1	1	FULL/DR	00:00:33	10.0.0.1
GigabitEthernet0/0				

Router#SS

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Static requires manual configuration of each route on all nodes, while OSPF automatically propagates routes throughout the network when only neighboring subnets are specified.

5.5 Testing Connectivity

The image displays two screenshots of the Cisco Packet Tracer interface, specifically the Command Prompt windows for PC0 and PC3. Both windows show the results of a ping command, indicating successful connectivity with 0% loss.

PC0 Command Prompt:

```
Physical Config Desktop Programming Attributes
Command Prompt
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time<1ms TTL=126
Reply from 192.168.5.2: bytes=32 time<1ms TTL=126
Reply from 192.168.5.2: bytes=32 time<1ms TTL=126
Reply from 192.168.5.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC3 Command Prompt:

```
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125
Reply from 192.168.3.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```


Questions:

1. How does a default static route differ from a manually configured static route to a specific network?

A specific static route defines a path to one particular network, while a default route handles all traffic for destinations not found in the routing table.

2. How does a router decide which route to use if multiple static routes exist?

The router selects the route with the longest prefix match to forward the packet.

3. What command would you use to remove a static route from a router?

Use the `no ip route` command followed by the exact network, mask, and next-hop address used during configuration.

4. How does the output of `show ip route` help in verifying static route configurations?

It displays installed routes, where static routes are marked with an "S" and the default route is marked with "S".

5. What is the main difference between static routing and OSPF?

Static routing requires manual entry for every path, while OSPF automatically discovers neighbors and updates the routing table dynamically.

6. What does the "O" symbol in the routing table represent?

The "O" symbol indicates that the route was learned through the OSPF dynamic routing protocol.

7. Why do we use wildcard masks in OSPF instead of subnet masks?

Wildcard masks are used to precisely define which interface IP addresses or ranges should participate in the OSPF process.

8. What is the purpose of Area 0 in OSPF?

Area 0 is the mandatory backbone area that connects all other OSPF areas to allow inter-area communication.

9. What happens if one link between routers goes down in OSPF?

OSPF automatically detects the failure and recalculates a new best path using the Link State Advertisement updates.

10. In which cases would you prefer to use static routing, and when would you choose dynamic routing?

Use static routing for small, stable networks or stub links, and dynamic routing for large, complex networks that require automatic failover.